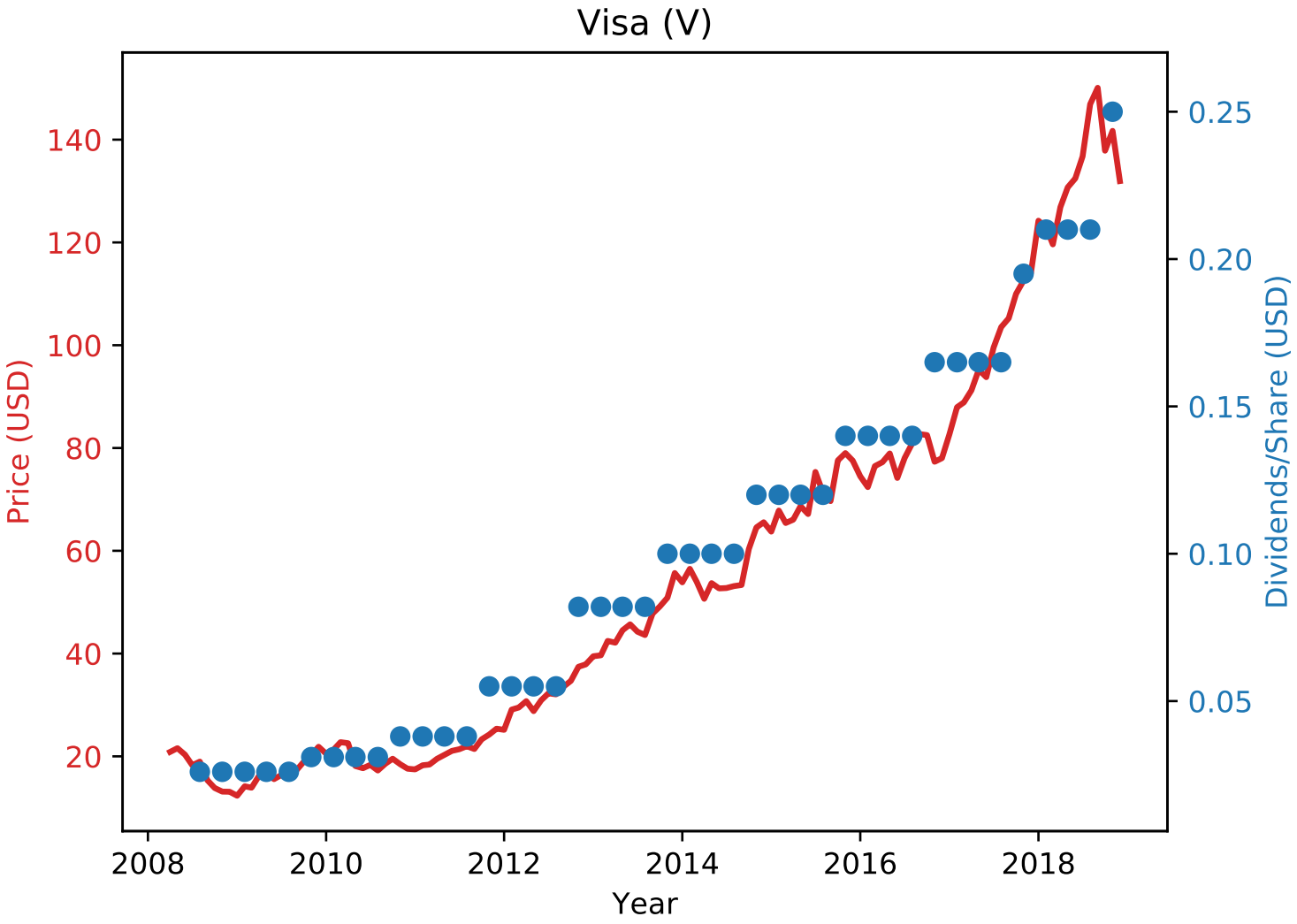
# Lecture 3.3: Stock Prices, Dividends, and Returns

# Stocks Prices, Dividends, and Returns

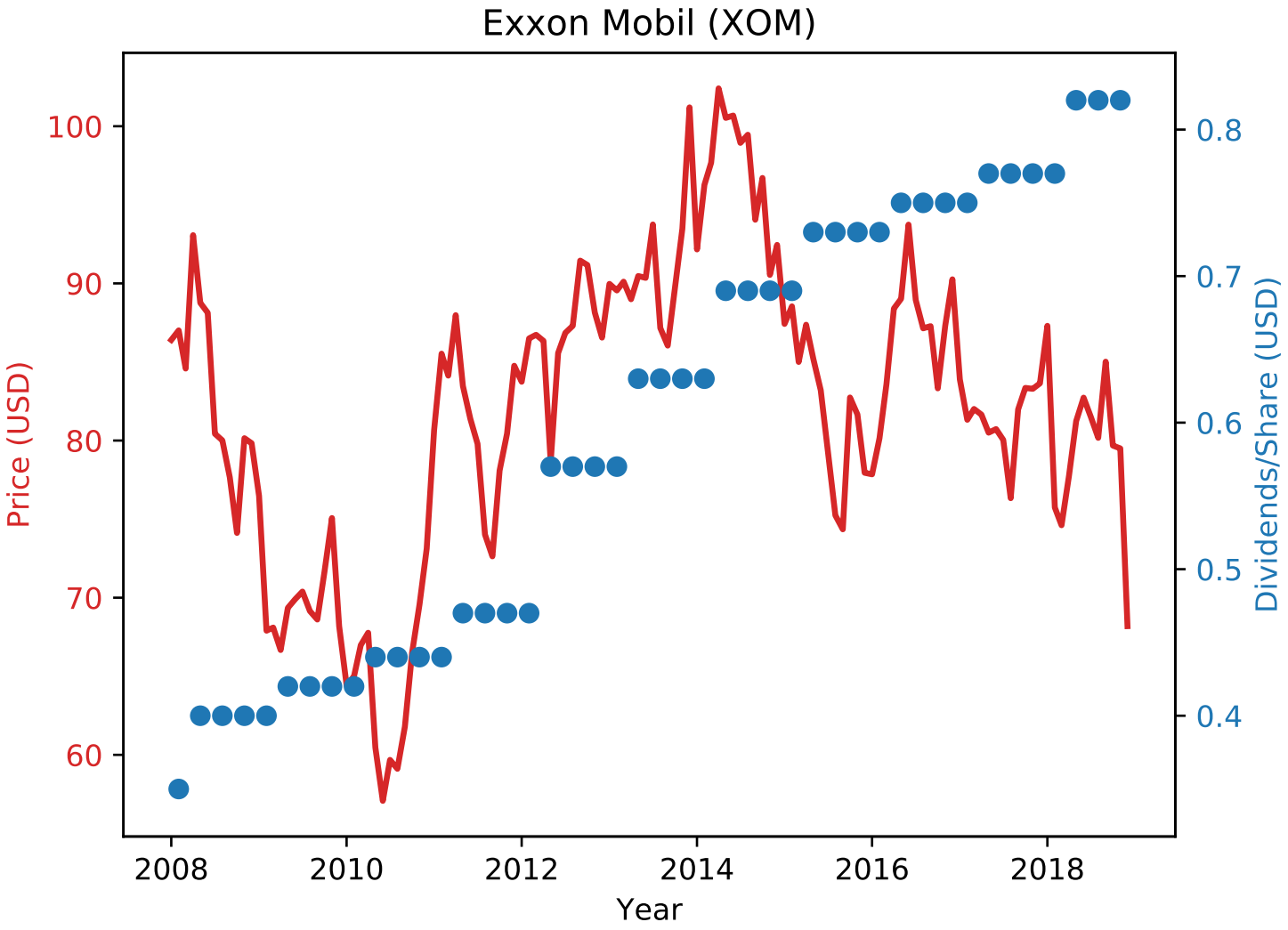
- The CAPM gives us a way of computing a stock's return
- We now want to compute prices
- 
- We will look at
  1. Discounted Cash Flows (DCF) Valuation
  2. Price-to-Earnings (P/E) Valuation

# Observations

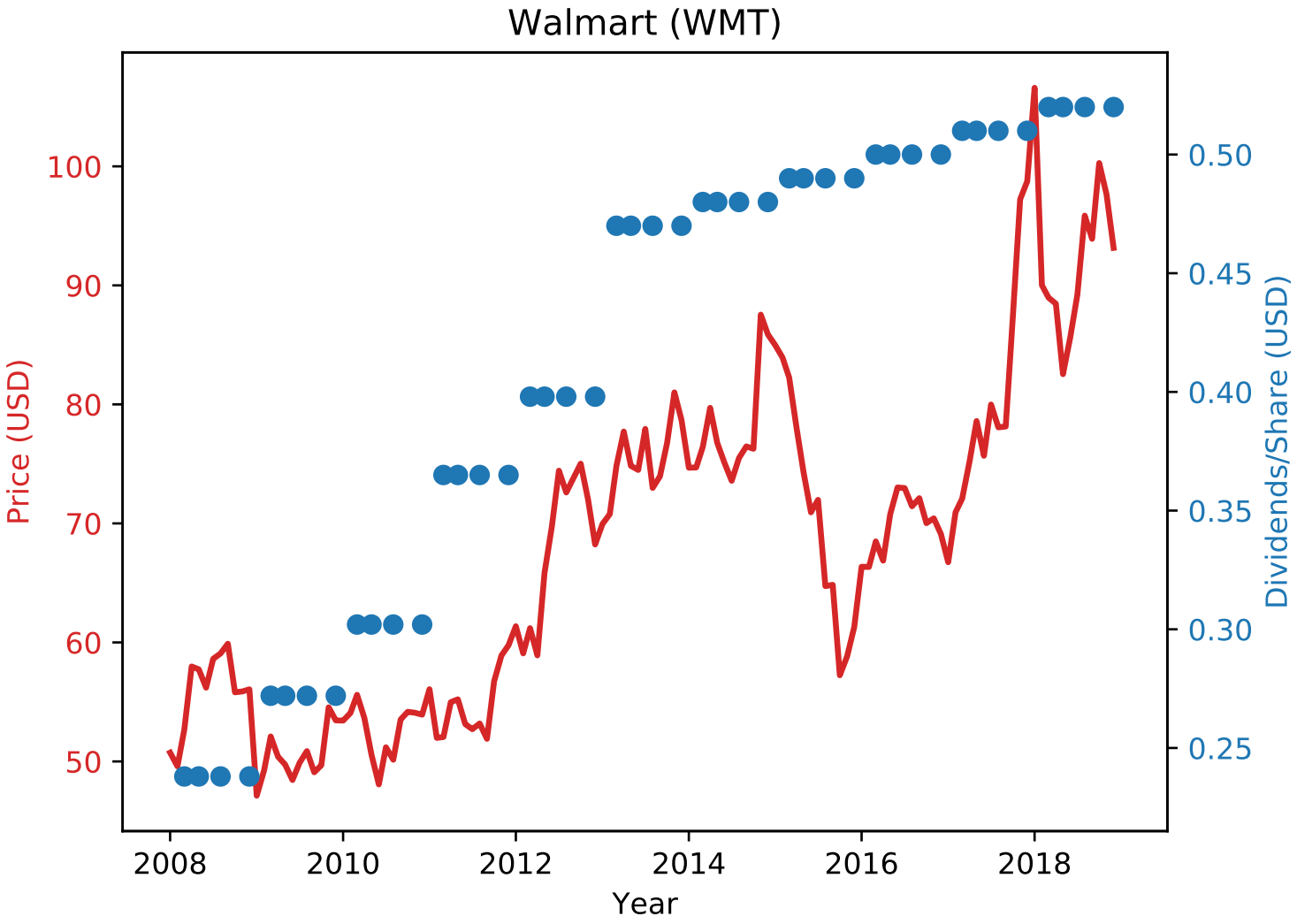
# Visa


Visa (V)
Price (USD)
Dividends/Share (USD)
Year
140
120
100
80
60
40
20
0.25
0.20
0.15
0.10
0.05
2008
2010
2012
2014
2016
2018

# Exxon


Exxon Mobil (XOM)
Price (USD)
100
90
80
70
60
Dividends/Share (USD)
0.8
0.7
0.6
0.5
0.4
2008
2010
2012
2014
2016
2018
Year

# Walmart


Walmart (WMT)
Price (USD)
100
90
80
70
60
50
Dividends/Share (USD)
0.50
0.45
0.40
0.35
0.30
0.25
2008
2010
2012
2014
2016
2018
Year

# Observations

- Verizon's dividends are (generally) constant across the four quarters in each year, but increase from the fourth quarter of one year to the first quarter of the next
- Exxon's dividends follow a similar pattern to those of Verizon, despite the much greater volatility in its stock price
- Walmart's dividends follow a similar pattern to those of Verizon and Exxon, but the annual growth in the dividend rate seems to kink in 2013

# DCF Valuation

# Stocks

- $P_1$ is the expected price in one period
- $r_S$ is the required total rate of return
- $D_1$ is the expected dividend
- The price should be

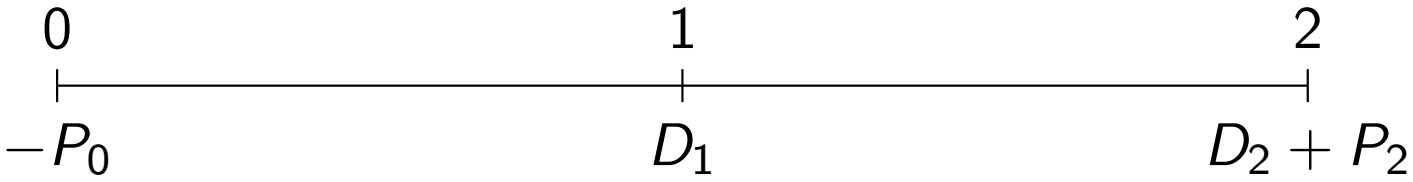
$$P_0 \overset{\text{should}}{=} \frac{D_1 + P_1}{1 + r_S}.$$

# Didend Yields, Capital Gains, and Total Returns

$$r \overset{\text{should}}{=} \frac{D_1 + P_1}{P_0} - 1 = \underbrace{\frac{D_1}{P_0}}_{\text{Didend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

- The <u>dividend yield</u> is the expected annual dividend of the stock divided by its current price
- The <u>capital gain rate</u> is the percent by which the stock price increases from the current period to the next
- The <u>total return</u> (or <u>holding period return</u>) is the sum of the dividend yield and the capital gain rate

# Stocks




- The prices should be

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r_S} + \frac{D_2+P_2}{(1+r_S)^2} \text{ and } P_1 \overset{\text{should}}{=} \frac{D_2+P_2}{1+r_S}.$$

- Does the two-year investor value the stock differently than the one-ear investor? No:

$$P_0 = \frac{D_1}{1+r_S} + \frac{D_2+P_2}{(1+r_S)^2} = \frac{D_1}{1+r_S} + \frac{P_1}{1+r_S} = \frac{D_1+P_1}{1+r_S}$$

# The Dividend Discount Model

- It doesn't matter whether the investor wants to hold the stock for one quarter, two quarters, or infinitely many quarters: she is always willing to pay the same price $P_0$
- In particular, the investor who holds the stock for infinitely many periods is willing to pay

$$P_0 \overset{\text{should}}{=} \frac{D_1}{1+r_S} + \frac{D_2}{(1+r_S)^2} + \frac{D_3}{(1+r_S)^3} + \cdots$$

  for the stock
- Implication: we don't need to know the prices $P_1$, $P_2$, ..., just the dividends $D_1$, $D_2$, $D_3$, ...

# Constant Dividend Growth

Suppose dividends grow at a constant rate $g$

| 0 | 1 | 2 | 3 | $\cdots$ |
|---|---|---|---|---|
| $-P_0$ | $(1+g)D_0$ | $(1+g)^2 D_0$ | $(1+g)^3 D_0$ | $\cdots$ |

where $D_0$ is the dividend that was just paid

$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{(1+r_S)} + \frac{(1+g)^2 D_0}{(1+r_S)^2} + \frac{(1+g)^3 D_0}{(1+r_S)^3} + \cdots = \frac{(1+g)D_0}{r_S - g}$$

# Constant Dividend Growth

- The Dividend Discount Model says that

$$P_0 \overset{\text{should}}{=} \frac{(1+g)D_0}{r_S - g} = \frac{D_1}{r_S - g}$$

- In this case, the <u>growth rate</u> $g$ is the <u>capital gain rate</u>:

$$\overbrace{\frac{D_1}{P_0} + g}^{\text{Specific Formula}} \overset{\text{should}}{=} r_S \overset{\text{should}}{=} \overbrace{\underbrace{\frac{D_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}}^{\text{General Formula}}$$

# P/E Valuation

# Price-to-Earnings (P/E) Ratios

- Unfortunately, computing prices via DCF can be extremely complicated and prone to error as one must be able to forecast the firm's payout policies, how those payouts correlate with the performance of the market, and so on
- In practice, finance professionals often use <u>multiples</u> to price stocks (or entire companies)

# Price-to-Earnings (P/E) Ratios

- Suppose that companies $A$ and $B$ have similar risk ($\beta_A \approx \beta_B$) and hence similar expected returns ($\mathrm{E}[r_A] \approx \mathrm{E}[r_B]$)
- P/E valuation assumes that earnings equal dividends: $E_A = D_A$ and $E_B = D_B$.
- Then $A$ and $B$ should have the same P/E ratio:

$$\frac{P_A}{E_A} \approx \frac{P_B}{E_B} \tag{1}$$

- If we know $B$'s P/E ratio and $A$'s earnings-per-share, we can approximate $A$'s price:

$$P_A \approx E_A \times \frac{P_B}{E_B} \tag{2}$$

- In this case, $P_B/E_B$ is the *multiple*

# Price-to-Earnings (P/E) Ratios

- Suppose that we want to price Honeywell (HON), which has a beta of 1.08. Lockheed Martin (LMT) has a beta of 0.93, while Walmart has a beta of 0.29. We should use the multiple of which company?

# Price-to-Earnings (P/E) Ratios

- Suppose that we want to price Honeywell (HON), which has a beta of 1.08. Lockheed Martin (LMT) has a beta of 0.93, while Walmart has a beta of 0.29. We should use the multiple of which company?
- Lockheed Martin. It's beta (and hence expected return) is much closer to that of Honeywell than that of Walmart.

# Price-to-Earnings (P/E) Ratios

| **Stock** | **Beta** | **P** | **E**(PS) | **P/E** |
|---|---|---|---|---|
| Honeywell | | | | 8.14 |
| Lockheed | | 386.70 | 22.82 | 16.95 |

- What should Honeywell trade for?

# Price-to-Earnings (P/E) Ratios

| **Stock** | **Beta** | **P** | **E**(PS) | **P/E** |
|---|---|---|---|---|
| Honeywell | | | | 8.14 |
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- What should Honeywell trade for?
- $$P_{\text{HON}} \approx E_{\text{HON}} \times \frac{P_{\text{LMT}}}{E_{\text{LMT}}} = 8.14 \text{ USD} \times \frac{386.70 \text{ USD}}{22.82 \text{ USD}} = \quad (3)$$
- Honeywell was *actually* trading for 161.49

# Notes to self

The SDF $m(1+r_f) = 1 - \mathrm{SR}[r_m]z_m$ prices the return of the total endowment, $r_m$, and the risk-free rate, $r_f$, where $z_m = (r_m - \mathrm{E}[r_m])/\mathrm{SD}[r_m]$. So if the stock's dividends grow at a random rate of $g$, then

$$
\begin{aligned}
P_0 &= \sum_{k=1}^{\infty} \mathrm{E}\left[(m(1+g))^k\right] D_0 && (4)\\
&\geq \sum_{k=1}^{\infty} \mathrm{E}[(m(1+g))]^k D_0 && (5)\\
&= \sum_{k=1}^{\infty} (1+r_f)^{-k} \mathrm{E}\left[(1-\mathrm{SR}[r_m]z_m)(1+g)\right]^k D_0 && (6)\\
&= \sum_{k=1}^{\infty} (1+r_f)^{-k} \left(1+\mathrm{E}[g]-\beta_g \mathrm{E}[r_m - r_f]\right)^k D_0 && (7)\\
&= \frac{D_1}{\beta_g \mathrm{E}[r_m - r_f] - \mathrm{E}[g - r_f]} && (8)
\end{aligned}
$$

where $D_1 = (1+\mathrm{E}[g]-\beta_g \mathrm{E}[r_m - r_f])D_0$